Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Tutorial 4 **Date: Aug 31, 2023**

1. Refer to the electric circuit shown in Figure 1. Evaluate V_{OC}, I_{SC} and R_{Th}

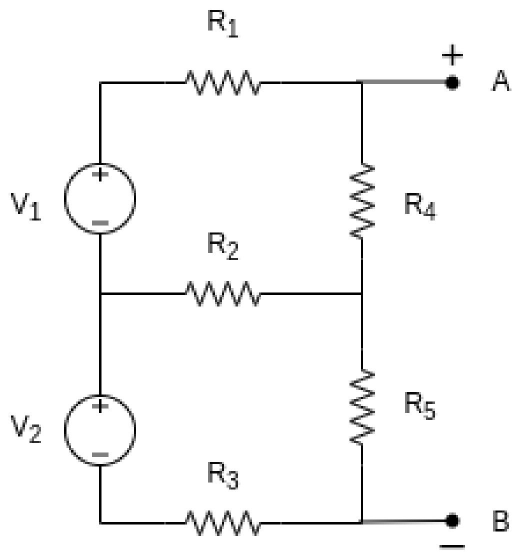

Figure 1: The electric circuit has two independent sources $V_1 = V_2 = 120V$ and the resistances $R_1 = R_2 = R_3 = 0.1\Omega$, $R_4 = 9.6\Omega$ and $R_5 = 5.76\Omega$

2. Refer to the electric circuit shown in Figure 2. Evaluate V_{OC}, I_{SC} and R_{Th}.

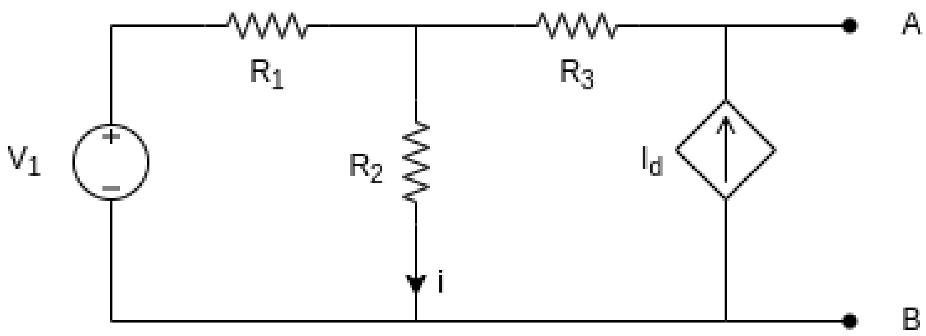

Figure 2: : The electric circuit with independent voltage source $V_1 = 20V$, dependent current source $I_d = ki = 1.5i$ and resistances $R_1 = 40\Omega$, $R_2 = 200\Omega$ and $R_3 = 100\Omega$

3. Consider the Boolean function

$$F(A, B, C, D) = \Sigma(1, 3, 4, 5, 6, 9, 11, 12, 14, 15).$$

   (a) Draw the Karnaugh map for the function F.
   (b) Find all Boolean expressions for F in the minimal SOP form.
   (c) Find all Boolean expressions for F in the minimal POS form.

4. Consider the K-map of the function $G(A,B,C,D)$ shown below in the figure below, with don't cares marked with the symbol ϕ. The designer can arbitrarily assign 0 or 1 to each location marked by ϕ (independently of each other). For example, the designer could for convenience choose the locations 2 and 7 to be 1's and choose locations 8 and 13 to be 0's resulting in a particular choice for the function $G(A,B,C,D) = \sum(0,1,2,3,4,5,7,12)$.

   (a) Find all Boolean expressions for G in the minimal SOP form.

   (b) Find all Boolean expressions for G in the minimal POS form.

CD \ AB	00	01	11	10
00	1	1	1	ϕ
01	1	1	ϕ	0
11	1	ϕ	0	0
10	ϕ	0	0	0